

Overview

In this experiment, morphological paradigms define the sets of grammatical forms that the classifier aims to recognize or distinguish. Each paradigm corresponds to a specific morphological or functional domain (e.g., tense/aspect markers, auxiliaries, modals), and determines the structure of the classification problem:

- **Multiclass required** paradigms (e.g., English {past, present, infinitive, modal}) train the model to distinguish which form of the paradigm is realized. Each token *must* realize one and only one of these gram components.
- **Binary** paradigms (e.g., {partizip}) test whether a particular component is present or absent.
- **Multiclass optional** paradigms (e.g., {yi, yijing, zaojiu, zaoyi}) include the possibility of no marking, allowing the model to learn when overt morphology is omitted.

Together, these paradigms operationalize the linguistic contrasts that the classifiers are trained to capture, translating the abstract structure of each language's TAM system into machine-interpretable prediction tasks.

Each language-specific section below has the following sections:

- Gram Components: a list of the discrete gram components that appear in the data
- Occurrence Patterns: a general overview of the occurrence patterns of the gram components
- Paradigms: sets of related grammatical forms that group together the possible realizations of a construction in each language
 - List: the paradigms themselves
 - Key: the "conceptual domain" of each paradigm (informal)

English

Gram Components

- Present (pres): includes present tensed verb
- Past (past): includes past tensed verb
- Participle (ptcp): includes a past passive participle
- Progressive (prog): includes a verb marked with progressive -ing
- Infinitive (inf): includes a non-tensed verb with infinitival to
- Modal: includes a modal, i.e., *could, may, might, should, will, would*

Occurrence Patterns

- Present, Past, Infinitive, and Modal cannot co-occur (complementary distribution)
- Modals cannot co-occur with each other
- Progressive can co-occur with anything else
- Participle can co-occur with anything else

Paradigms

List

- P1: [ptcp] (binary)
- P2: [pres, past, inf, modal] (multiclass required)
- P2a: [can, could, may, might, must, ought_to, shall, should, will, would] (multiclass optional)
- P3: [prog] (binary)

Key

- P1 = Complex/Simplex
- P2 = TAM Root
- P2a = Lexical Modals
- P3 = Progressive Aspect

German

Gram Components

- | | | |
|--------------------------|-------------------------|------------------|
| • Infinitiv (infinitiv): | • Konjunktiv I (kon1): | • <i>Moegen</i> |
| • Präsens (prasens): | • Konjunktiv II (kon2): | • <i>Muessen</i> |
| • Präteritum (prat): | • Partizip (partizip): | • <i>Sollen</i> |
| • <i>Werden</i> | • <i>Duerfen</i> | • <i>Wollen</i> |
| • Indikativ (indikativ): | • <i>Koennen</i> | • <i>Haben</i> |
| | | • <i>Sein</i> |

Occurrence Patterns

- Präsens, Präteritum, and Infinitiv cannot co-occur
- Konjunktiv I, Konjunktiv II, and Indikativ cannot co-occur
- Modals can occur with Partizip
- Modals can (must!) occur with tense, either Präsens or Präteritum
- *Werden* is weird: an analytic tense that can co-occur with everything tense can, but also including tense itself
- *Haben* and *sein* cannot co-occur

Gram Paradigms

List

- P1: [partizip] (binary)
- P2: [infinitv, prasens, prat] (multiclass required)
- P3: [indikativ, kon1, kon2] (multiclass required)
- P4: [modal] (binary)
 - P4_a: [duerfen, koennen, moegen, muessen, sollen, wollen] (multiclass optional)
- P5: [werden] (binary)
- P6: [haben, sein] (multiclass optional)
- P7: TBD, consider specific derived/multicomponent oppositions? Werden?

Key

- P1 = Complex/Simplex
- P2 = Temporal Root
- P3 = Mood
- P4 = Modal
 - P4_a = Lexical Modals
- P5 = Werden
- P6 = Auxiliaries
- P7 = TBD

Mandarin

Gram Components

- Aspectual particles: *le, guo, no_asp* (all tokens that have no aspectual particle)
- "Auxiliary have": *you*
- Resultative RC marker: *suo_de*
- Temporal adverbs: *yi, yijing, zaojiu, zaoyi*
- Aspectual compounds (e.g., *qilai*, those including *dao*, etc.)
- Nullmarked

Occurrence Patterns

- Aspectual particles can co-occur with adverbs, *you*, and/or compounds, but not with each other
- "Auxiliary have" can co-occur with aspectual participles (and maybe *suo_de*)
- *Suo_de* can co-occur with adverbs, (and maybe *guo, you*)
- Adverbs can co-occur with anything (but presumably not each other)
- Compounds can co-occur with anything
- Nullmarked is exclusive (by definition cannot co-occur with anything)

Gram Paradigms

List

- P1: [*le, guo, no_asp*] (multiclass required)
- P2: [*adverbial*] (binary)
 - P2_a: [*yi, yijing, zaojiu, zaoyi*] (multiclass optional)
- P3: [*suo_de*] (binary)
- P4: [*compound*] (binary)
- P5: [*you*] (binary)
- P6: [*nullmarked*] (binary)

Key

- P1 = Aspectual Particles
- P2 = Adverbial
 - P2_a = Individual Adverbials
- P3 = Resultative RC

- P4 = Aspectual Compounds
 - P5 = Auxiliary
 - P6 = Nullmarked
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Japanese

Gram Components

- Verb "tense": nonpast, *te*, *ta*
- Completive: (te) *shimau*
- Auxiliaries: *aru*, *iru*
- Nominalizer: *koto_ga*
- Recent/immediate past: (-*ta*) *tokoro_da*, *bakari_da*

Occurrence Patterns

- *ta* can co-occur with anything except nonpast and *te*
- *te* can co-occur with *iru*, *shimau*
- *shimau* can co-occur with (preceding) *te* and (following) *ta*
- *iru* can co-occur with *te*
- *aru* can co-occur with *ta*, *shimau*, *koto_ga*, and recent/immediate past

Gram Paradigms

List

P1: [ta, te, nonpast] (multiclass required)
 P2: [iru] (binary)
 P3: [shimau] (binary)
 P4: [koto_ga (aru)] (binary)
 P5: [bakari, tokoro] (multiclass optional)
 P6: [aru] (binary)

Key

P1 = Temporal Anchor
 P2 = Progressive Resultative
 P3 = Completive
 P4 = Experiential
 P5 = Recent Past
 P6 = Existential